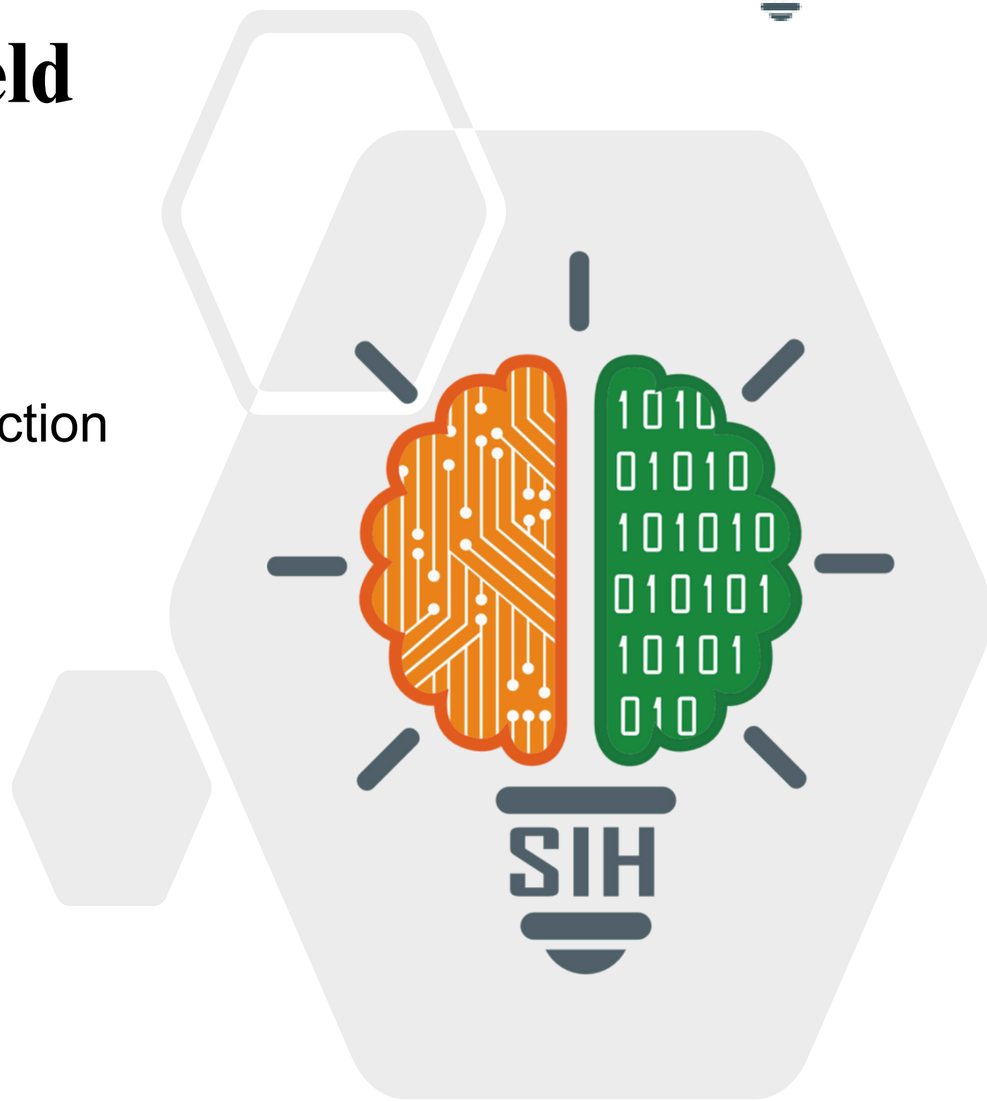
SMART INDIA HACKATHON 2025

Team- **Rock_Shield**

- **Problem Statement ID -** 25071
- **Problem Statement Title -** AI-Based Rockfall Prediction and Alert System for Open-Pit Mines
- **Theme -** Disaster Management
- **PS Category -** Software
- **Team ID -** 77901
- **Team Name -** Rock_Shield



Rock_Shield

SMART ROCK

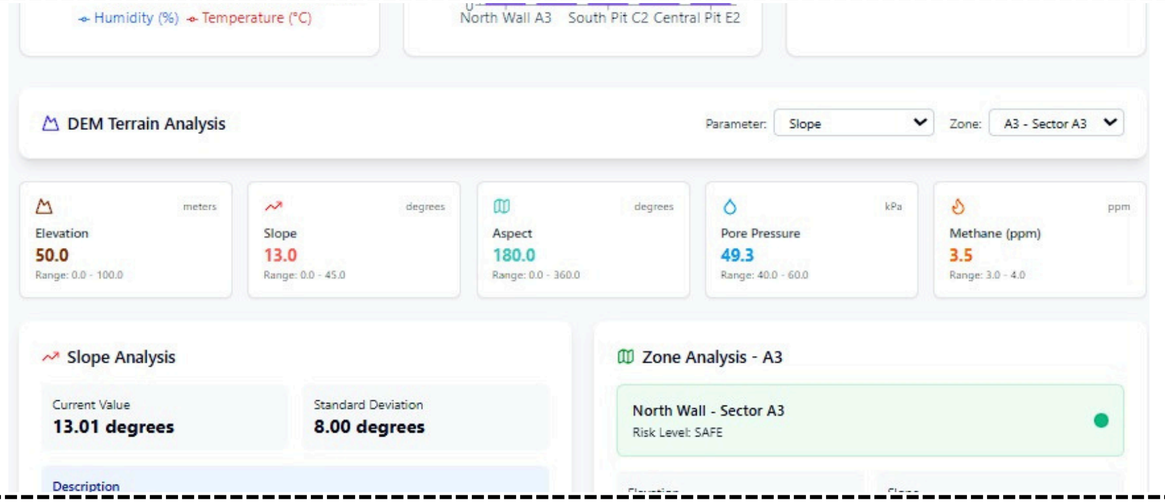
SMART INDIA HACKATHON 2025

Proposed Solution

- **AI Prediction:** LSTM + XGBoost ensemble forecasts rockfall risk ahead of time with interpretable outputs.
- **Synthetic Data:** DEM-based features and simulated geotechnical/weather patterns overcome limited real datasets.
- **IoT Fusion**: Integrates with existing mine instruments; LoRa ESP32 nodes fill data gaps where needed.
- **Alerts & UI**: Role-based dashboard for engineers and on-site siren/LED alerts for workers to avoid confusion.
- **Ops Model**: Hybrid edge system with minimal connectivity needs, scalable to mines, tunnels, quarries, and dams.

Unique Value Proposition

- **Predictive Safety:** Moves from reactive alarms to proactive rockfall forecasting.
- **Cost-Disruptive:** Affordable IoT and Jetson-based system vs multi-crore radar.
- **Hardware-Agnostic**: Works with existing mine sensors; expandable if needed.
- **No Data Monopoly**: Synthetic DEM + simulated datasets remove dependency barriers.
- **Controlled Access**: Dashboards only for engineers; workers receive siren/LED alerts.



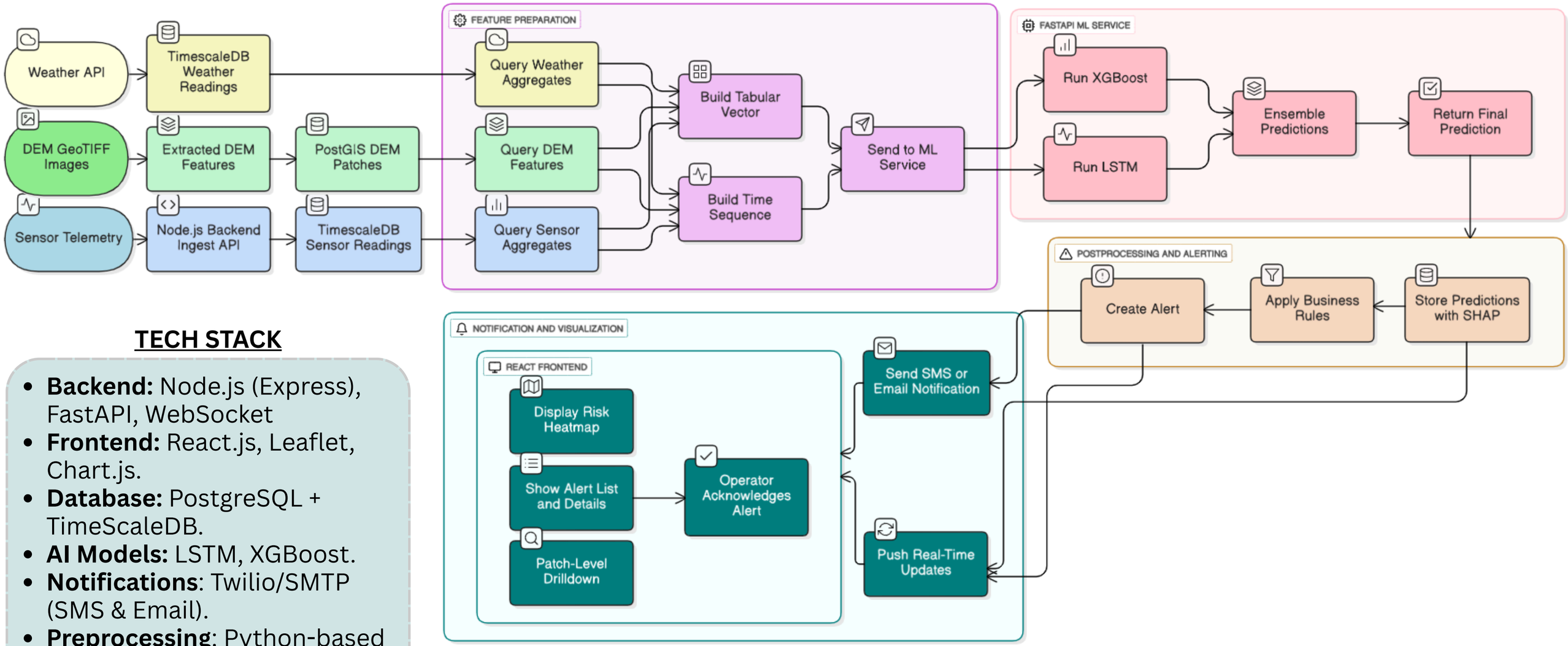
Key Characteristics

- **Relevance**: Targets real mining risks—human safety, equipment damage, and downtime.
- **Accurate Dashboards:** Real-time alerts, visual trends, and clear insights.
  - **Multi-Source Fusion**: Drone Imagery-based DEM, IoT data, and weather inputs enhance prediction reliability.

Rock_Shield

TECHNICAL APPROACH

SMART INDIA HACKATHON 2025



TECH STACK

- **Backend:** Node.js (Express), FastAPI, WebSocket
- **Frontend:** React.js, Leaflet, Chart.js.
- **Database:** PostgreSQL + TimeScaleDB.
- **AI Models:** LSTM, XGBoost.
- **Notifications**: Twilio/SMTP (SMS & Email).
- **Preprocessing**: Python-based tools & simulations.

Rock_Shield

FEASIBILITY AND VIABILITY

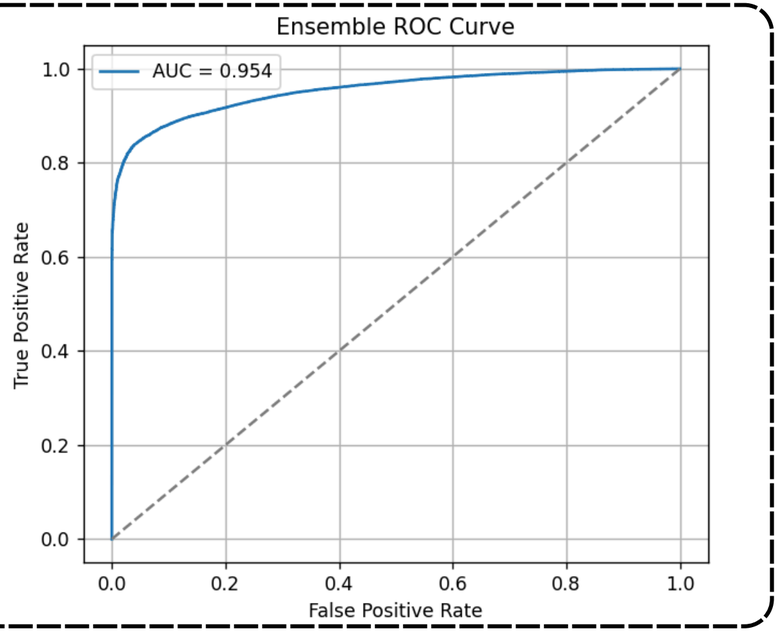
SMART INDIA HACKATHON 2025

Challenges

- **Data Scarcity**: Limited labeled datasets for rockfall prediction.
- **Sensor Noise:** Commodity sensors are prone to fluctuations & false positives.
- **Connectivity Gaps:** Remote mines frequently lack stable internet/network connections.

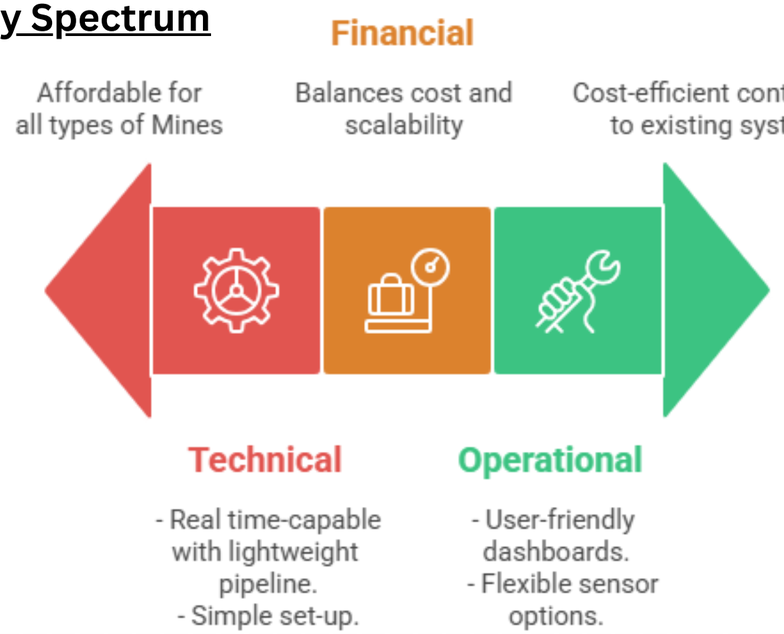
Solutions

- **Synthetic Training Data:** Generate samples via DEM + weather APIs and considering factor of safety.
- **Signal Validation:** Filters, rolling averages & multi-sensor cross-checks.
- **Edge-based Architecture:** Edge processing for instant alerts + future scope with Jetson Nano.



The ensemble ROC curve shows how well our model separates safe vs risky conditions. With an AUC of 0.954, the system demonstrates high reliability and low false alarms, making it suitable for real-world deployment.

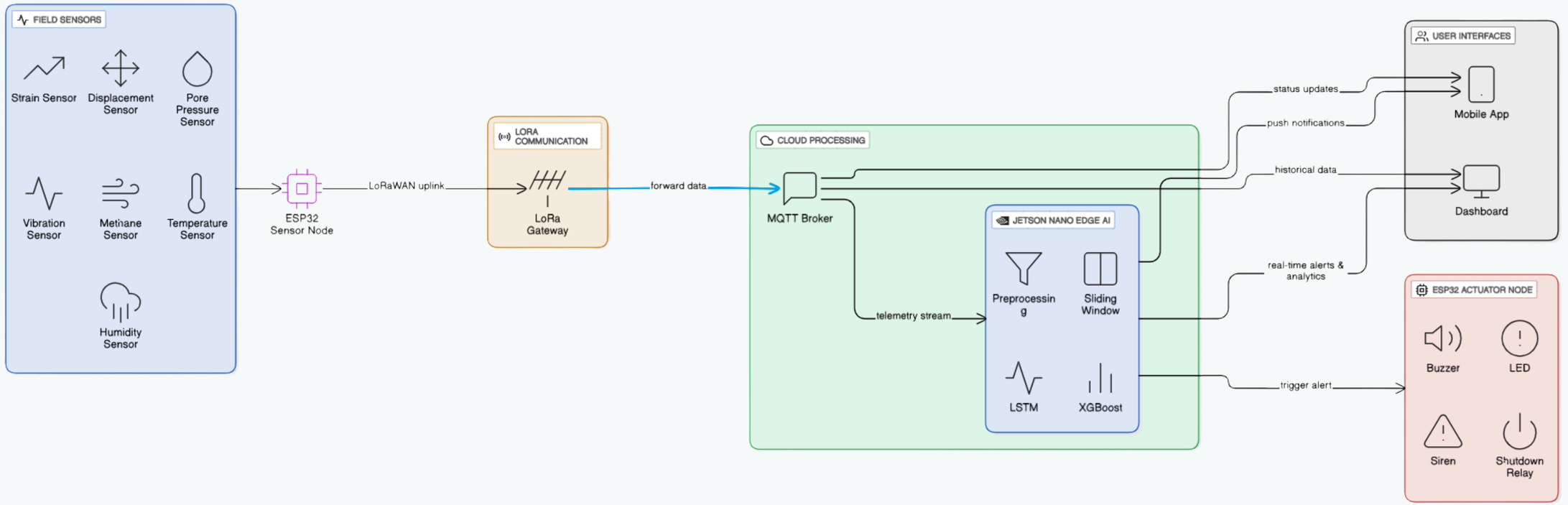
Feasibillity Spectrum



Rock_Shield

FUTURE SCOPE

SMART INDIA HACKATHON 2025

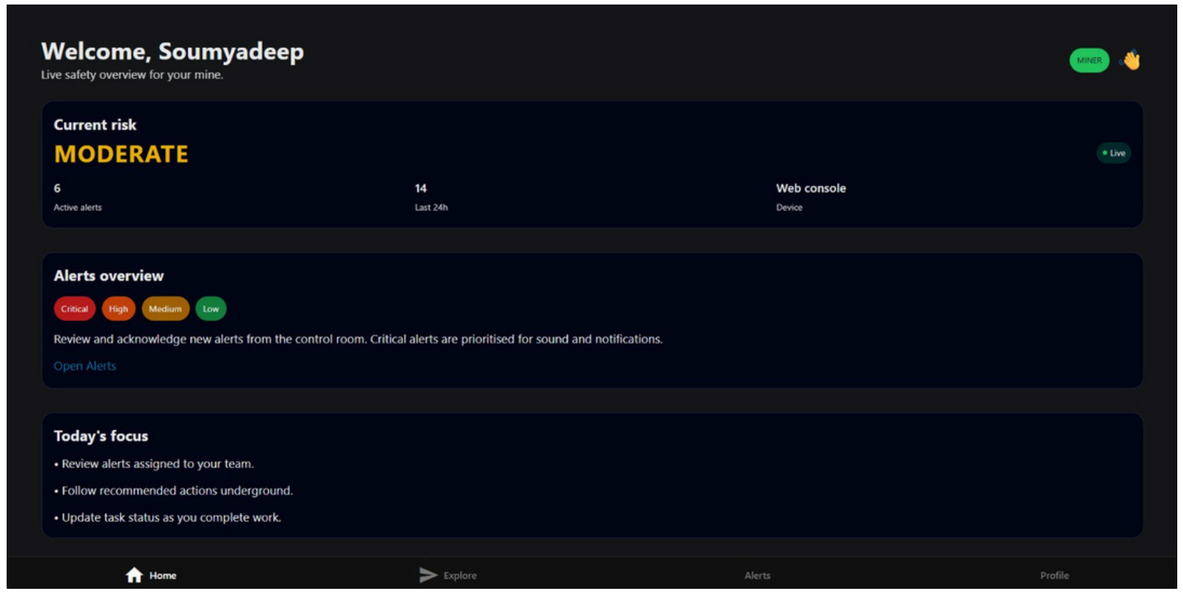


The future scope is to turn this into a complete IoT + Edge AI safety system. Sensors send data to an ESP32 node, which transmits it via LoRa to a gateway and MQTT broker. A Jetson Nano processes the data, runs the LSTM + XGBoost model, and generates real-time alerts. These alerts activate an ESP32 actuator node to trigger buzzers, LEDs, or shutdowns, while a dashboard (mobile + web) displays live alerts and historical trends. This creates a scalable, automated system for real-time monitoring and rapid response.

Rock_Shield

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2025



Application User Interface

References

- 3d Mapping Datasets: https://data.mendeley.com/datasets/c5s76mj4bm/5
- Rockfall Dynamics Prediction Using Data-Driven Approaches: A Lab-Scale Study (MDPI — Geotechnics, Ghahramanieisalou & Sattarvand)
- Rockfall detection using LiDAR and deep learning (Farmakis et al., Engineering Geology 2022)
- Machine learning prediction of landslide deformation behaviour using acoustic emission and rainfall measurements (Engineering Geology)

Research

Our research examines how different geotechnical sensors—such as tilt, displacement, pore pressure, vibration, soil moisture, and rainfall—capture the physical precursors of slope failure and how these signals can be combined to estimate the probability of rockfall. By analyzing their influence on slope stability and integrating them into a probabilistic framework, we translate raw sensor readings into actionable risk forecasts and show that on our dashboard.